

Assignment -1

Module-1

1. Explain the following with syntax and an example.
i) for loop ii) while loop iii) break iv) continue
2. Explain how to define a function and to make a function call by passing an argument with an example.
3. Explain the salient features of Python.
4. Develop a python program to read 2 numbers from the keyboard and perform the basic arithmetic operations based on the choice. (1-Add, 2-Subtract, 3-Multiply, 4-Divide).
5. Develop a program to print the name and year of birth of a person. Display whether the person is a senior citizen or not.
6. Develop a program to generate Fibonacci sequence of length N. Read N from the console.
7. Define debugging. Explain different types of errors-syntax errors, runtime errors, and semantic errors with suitable examples.
8. Define program? Explain the basic concepts of program.
9. Define variable. Explain the rules for naming the variable with an example.
10. Explain how iteration and conditional statements are used in Collatz $3n+1$ sequence implementation.
11. Explain the following functions with an example:
i) len() ii) int() iii) print() iv) input() v) str()
12. Solve the following expressions:
i) $10 + 6 \% 4 * 2$ ii) $3/2*4+3+(10/4)**3-2$ iii) $5 + 3 * 3 ** 2$ iv) $(-3) ** 2$
13. Explain return values in functions. Write a function that takes two numbers as arguments and returns their sum, difference, and product.
14. Write a simple program to print even numbers up to n.
15. List and explain math operators used in Python with example.